Enterprise Learning Platform

with Skill and Career Guidance System

Project Plan

Enterprise Learning Platform with Skill and Career Guidance System

Team C

Prepared for team review

1. Project Overview

The Enterprise Learning Platform with Skill and Career Guidance System is a full-stack online learning platform where students browse and enroll in courses, learn through structured lessons, take quizzes and assignments, track their progress, and earn badges and certificates. Instructors manage their own courses and students, and admins oversee the entire platform. The project follows a client-server architecture: a React single-page application (the client) talks to a Java Spring Boot REST API (the server), which stores data in a MySQL database. A dedicated AI layer adds a chatbot, a personalized learning roadmap, and skill-gap analysis, giving the platform its career-guidance identity rather than being a plain course-delivery tool.

2. Objectives

- Deliver a secure, working authentication system (Google OAuth2 + JWT) as the foundation of the platform.
- Build a real, database-backed learning experience: courses, lessons, quizzes, and progress tracking.
- Extend the platform with engagement and career-facing tools: assignments, certificates, resume building, community forum, and gamification.
- Layer AI-powered guidance on top of the core platform, and provide dedicated tooling for instructors and administrators.
- Reach a production-ready state: secrets rotated, cloud database in place, and access control fully enforced.

3. Team & Roles

Member	Role	Responsibility
Nandini Verma	Frontend / UI Lead	Building all pages & components (React) — dashboards, course pages, forms, layouts
Rahul	Backend / API Developer	Spring Boot REST APIs, controllers, services, business logic
Rajeev	Database Specialist	MySQL schema design, entities/models, repositories, data relationships
Sasi Madhuri	Authentication & Security Engineer	Login/signup, JWT, Google OAuth2, Spring Security setup
Madhumitha	AI Integration & Deployment	Gemini-powered AI features (chatbot, roadmap), testing, deployment/setup

4. Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite	Single-page application UI, fast dev/build tooling
Routing	React Router 7	Client-side page navigation
Frontend state	React Context API	Auth state, toast notifications, theme
Backend	Spring Boot 3.5 (Java 17)	REST API server
Security	Spring Security + JWT + OAuth2	Login, session tokens, Google sign-in
Database	MySQL	Persistent data storage
ORM	Spring Data JPA / Hibernate	Java-to-database mapping
Email	Spring Mail	Password reset emails, reminder notifications
AI	Gemini API	Chatbot, personalized roadmap, skill-gap analysis
Build tools	Maven (server), npm (client)	Dependency management & builds

5. Project Timeline — Milestone Breakdown

The project is delivered across four milestones, each building directly on the previous one rather than as parallel, independent workstreams.

Milestone	Focus	Est. Duration
Milestone 1	Foundation — Google OAuth2 + JWT authentication, MySQL users table, themed landing page	2 weeks
Milestone 2	Real dashboard, course catalog, lesson/quiz engine, forgot/reset password, profile settings	3 weeks
Milestone 3	Assignments, certificates, resume builder, internships, forum, leaderboard, gamification	3 weeks
Milestone 4	AI chatbot/roadmap/skill-gap, instructor & admin tools, production hardening	3 weeks

6. Key Deliverables by Milestone

Milestone 1

- Working Google OAuth2 login flow end to end
- JWT issuing, validation, and route protection
- MySQL users table auto-created via Hibernate

- Themed landing page (Hero, Routes, Instruments, TrailLog, Navbar, Connect)

Milestone 2

- Dashboard driven by real backend data (stats, streak, weekly chart, achievements)
- Course catalog with real enrollment
- Lesson player supporting video, reading, and server-graded quiz lessons
- Forgot/reset password via real email
- Account/profile settings page

Milestone 3

- Assignments and practice modules
- Certificates with public verification
- Resume builder
- Internships listing, live sessions, and schedule
- Forum, leaderboard, and expanded gamification
- Notifications, announcements, and analytics page

Milestone 4

- AI chatbot, personalized roadmap, and skill-gap analysis
- Instructor dashboard and course/student management
- Admin dashboard with platform-wide statistics
- Role-based access control finalized across student/instructor/admin
- Production-readiness checklist addressed (secret rotation, cloud DB, rate limiting)

7. Risks & Mitigation

Risk	Mitigation
Secrets (OAuth secret, JWT secret, mail credentials) exposed during development/debugging	Treat anything shared in chat, screenshots, or commits as compromised; rotate all secrets before production; keep application.properties out of version control
Google OAuth redirect URI mismatch blocking login	Keep the authorized redirect URI in Google Cloud Console in sync with the backend port at all times; document the exact value in setup instructions
Local-only MySQL creates a single point of failure and blocks team collaboration	Plan and execute a cloud MySQL migration before production use
Feature branches diverging from main for too long	Regular merges into main via reviewed Pull Requests rather than one large merge at the end

Risk	Mitigation
Dev-only endpoints (e.g. demo-data seeding) left accessible in production	Explicit production-readiness checklist item to remove or lock these down before launch
AI features relying on placeholder logic instead of a real provider	Verify AI Assistant Controller is wired to a live Gemini API key before considering Milestone 4 complete

8. Definition of Done

- Feature has been built, manually tested against the running backend (via Postman and the live browser), and pushed to the team repository.
- No hardcoded secrets are committed; configuration is read from environment variables or a git-ignored properties file.
- New backend endpoints follow the existing layered structure (Controller → Service → Repository → Model) and return the platform's standard response shape.
- New frontend pages are wired into routing and protected appropriately (public vs. authenticated vs. role-restricted).
- Known bugs discovered during the milestone are documented, not just silently fixed.

9. Tools & Workflow

- Version control: Git, hosted on GitHub, with feature branches merged into main via Pull Request
- Backend IDE: IntelliJ IDEA
- API testing: Postman
- Frontend dev server: Vite (npm run dev)
- Backend build: Maven wrapper (mvnw / mvnw spring-boot:run)
- Database: local MySQL during development, cloud MySQL (Aiven) planned for production